

If humans were truly rational, would supermarkets still exist in their current form?

Zijing Wu - Hong Kong

Introduction

The rational consumer, or *Homo Economicus*, is assumed to be a rational agent seeking to maximize utility subject to a budget constraint. In theory, supermarket customers should yield an accurate representation of this, precisely executing pre-planned choices based on their own needs. However, Bell, Corsten, and Knox (2010) found that 70% of purchasing decisions are made in-store rather than planned in advance, suggesting a deviation from rational choices, perhaps driven by the supermarket architecture itself. This essay establishes the formal rational benchmark, examines how supermarkets exploit deviations from it through various techniques, and constructs what the supermarket would look like if consumers were genuinely rational.

The Rational Consumer Benchmark

In Neoclassical Economics, the rational consumer is based on two important axioms - completeness and transitivity. Using the notation $\succeq$ to denote preference, a consumer is rational when his choices across two bundles of goods A and B satisfy the following properties (Mas-Colell, Whinston, and Green, 1995):

- Completeness - $A \succeq B$, $B \succeq A$, or $A \sim B$ (Indifferent); This ensures that consumers can rank any two bundles of goods.
- Transitivity - If $A \succeq B$ and $B \succeq C$, then $A \succeq C$; This ensures that the rankings are logically consistent.

In addition, we can use an utility function $U(x)$ to model the utility of a consumer, where x is the amount of a good consumed. The utility received from choosing a combination of two goods can therefore be modeled as $U(x_1, x_2)$, as seen as the curve in Fig.1. In addition, rational consumers also faces budget constraints, hence, the constrained maximum utility $\text{Max } U(x_1, x_2)$ occurs under the scenario where the cost m equals to the aggregate cost of good x_1 and x_2 , or $m = p_1 x_1 + p_2 x_2$. The budget constraint can be represented as a linear curve. Hence, for a rational consumer, their optimal choice would be at the point (x_1^*, x_2^*) .

In a supermarket, this implies the consumer would compare unit prices across all available substitutes before every purchase.

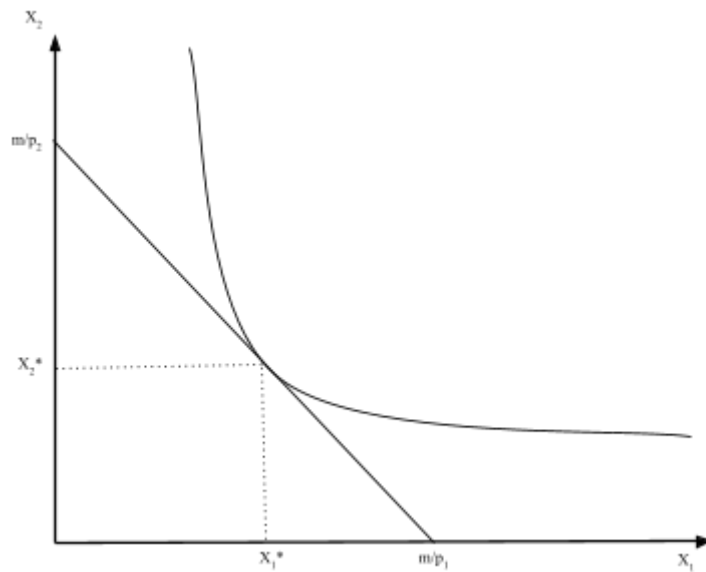


Fig.1 - Indifference Curve

The Architecture of Irrationality

Anchoring and Reference Dependence:

In reality, consumer decision-making often deviates from rational thinking due to cognitive biases, such as anchoring, the tendency to rely heavily on initial reference points to make judgments (Tversky and Kahneman, 1974). Supermarkets exploit this through price architecture, using an initial higher price to make the discounted price seem a gain rather than an absolute expenditure.

This behavior can be explained by the Prospect Theory (Kahneman and Tversky, 1979), which argues that individuals evaluate outcomes as gains or losses relative to a subjective reference point rather than absolute states of utility. Because the value function is steeper for losses than for gains (a phenomenon known as loss aversion)(See Fig 2), the perceived "saving" triggers a stronger psychological impulse than a rational evaluation of the product's marginal utility.

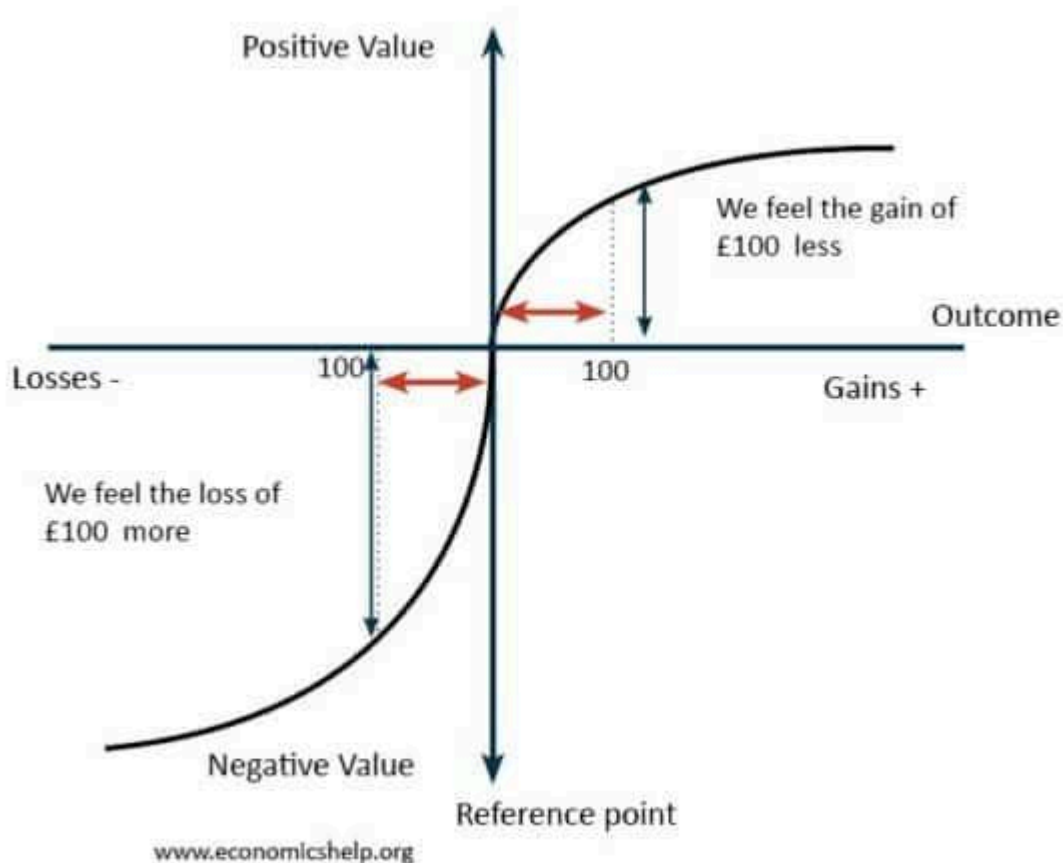


Fig 2. Prospect Theory (Pettinger, 2018)

Returning to the framework in Fig. 1, anchoring distorts the consumer's perceived budget constraint. When a higher "was" price is displayed alongside a sale price, the consumer perceives the budget line as having shifted outward, as though purchasing power has increased, even though the objective constraint $m = p_1x_1 + p_2x_2$ remains unchanged. This causes the consumer to select a bundle that deviates from (x_1^*, x_2^*) , overspending on the anchored goods. The CMA's (2024) investigation into loyalty pricing transparency confirms that regulators recognise this architecture as potentially harmful, an investigation entirely unnecessary if consumers were rational.

Choice Architecture and Placement Effects

In a neoclassical framework, a store's physical arrangement should be irrelevant to a rational consumer concerned only with utility and price. However, supermarkets utilize nudges, subtle environmental changes that alter behaviour without restricting options (Thaler and Sunstein, 2008). Product placement is a primary example: items at eye-level or end-of-aisle positions are made significantly more noticeable than those on lower shelves. This exploits the default effect (Thaler and Sunstein, 2008): consumers disproportionately select options made salient or easy to reach, treating prominence as an implicit recommendation. Research suggests

eye-level products receive approximately 35% more visual attention than those on lower shelves (Ebster and Garaus, 2015).

In terms of the formal model in Fig. 1, placement effects operate as though the perceived price of salient goods is lower than it actually is. By reducing the cognitive cost of selecting prominent items, the retailer effectively tilts the consumer's subjective budget line, shifting the chosen bundle away from (x_1^*, x_2^*) toward one that maximises store margins rather than consumer utility.

Counterarguments and Qualifications

Before constructing the rational supermarket, we must address whether the behaviours documented above genuinely represent irrationality, since if consumers are already rational, the supermarkets would remain the same.

One counterargument of consumer irrationality is the revealed preference theory (Samuelson, 1938), which posits that observed choices, by definition, reveal true preferences. A shopper selecting an eye-level product has simply revealed a preference for convenience. However, this defence is circular and unfalsifiable: any behaviour can be rationalised post hoc as a "revealed preference," rendering the theory immune to contradiction and therefore unscientific by Popperian standards (Popper, 1959).

Another challenge comes from Simon's (1955) concept of bounded rationality. This theory suggests that, given constraints to cognitive ability and time, relying on heuristics to reach a satisficing outcome rather than an optimal one is rational. In this view, the biases of supermarket consumers are not failures of reasoning but pragmatic adaptation to constraints. However, rather than defending the traditional rational model, Simon's (1955) concept of bounded rationality illustrates exactly why supermarket choice architecture proves so effective. Simon posited that consumers, constrained by cognitive limits and time, rely on heuristics to reach a satisficing outcome rather than an optimal one. Supermarkets exploit this exact pragmatic adaptation. It is precisely because consumers rely on these cognitive shortcuts that retailers can design environments to intercept these heuristics. For instance, placing high-margin goods at eye level capitalizes on these limitations. The consumer's shortcut does not serve their own utility maximization but is instead carefully redirected to serve the retailer's commercial interests.

The Rational Supermarket

If humans were to act truly rationally and adhere strictly to completeness and transitivity, the supermarket would be fundamentally transformed.

The most immediate change would be the product range. Decoy options, which serve to increase the value of adjacent products, would disappear completely, as rational agents

evaluate the marginal utility of each bundle independently and disregard any decoys. Instead of stocking thirty functionally equivalent pasta sauces, the rational supermarket would retain only those variants occupying a distinct position on the price-quality frontier.

Store layout would be dictated entirely by logistical efficiency. There would be no purpose to place essentials like milk or bread towards the back of the store, which was originally intended to maximize exposure to impulse purchases along the route. A rational consumer is immune to unplanned purchasing, so the optimal layout minimises transaction costs, specifically the time spent locating goods. Products would be arranged by category in a logical, clearly signposted sequence, closer to a warehouse than a supermarket. Shelf placement would become commercially irrelevant, since eye-level positioning influences only those susceptible to the default effect. Retailers would therefore allocate shelf space by storage efficiency or alphabetical order rather than by supplier payments for prominent positioning.

Price architecture would also collapse into full transparency. “Was/now” labels and other techniques designed to exploit cognitive biases would all become redundant. A rational consumer would evaluate prices solely based on its absolute price and their marginal utility, so pricing labels would be made to simply communicate the pricing clearly. Price comparison across substitutes, which the current supermarket deliberately makes difficult through inconsistent unit pricing and varying pack sizes, would need to be standardized, since any retailer that failed to do so would simply lose customers to one that did. Competition in this market reduces entirely to price and product quality, with no manipulation.

What remains, then, is not a supermarket in any recognisable sense but a distribution point, a sparse, unbranded warehouse optimised for speed and cost, or more likely, a delivery network rendering the store unnecessary altogether. The distance between this counterfactual and the supermarket as it currently exists is itself the strongest evidence that the industry is built not on serving rational preferences but on systematically exploiting their absence.

Conclusion

This essay has argued that the distance between the real supermarket and the rational one is the clearest measure of what the standard economic model leaves out. If consumers were truly rational, anchoring strategies would fail, shelf placement would be commercially worthless, and decoy pricing would be ignored. The supermarket would be transformed into a warehouse where the only competitive variable is price.

Behavioral economics does not merely identify isolated anomalies. It explains why an entire industry is structured the way it is. Every pound spent on choice architecture and manipulative pricing is a pound wagered on the prediction that consumers will not behave rationally. The supermarket's continued profitability suggests that the wager pays off.

Bibliography

A, S.P. (1938). A Note on the Pure Theory of Consumer's Behaviour. *Economica*, [online] 5(17), pp.61–71. doi:<https://doi.org/10.2307/2548836>.

Bell, D., Corsten, D. and Knox, G. (2010) Unplanned Buying on Shopping Trips. Marketing Science Institute Working Paper Series, Report No. 10-109.

Chernev, A., Böckenholt, U. and Goodman, J. (2015). Choice overload: A Conceptual Review and Meta-Analysis. *Journal of Consumer Psychology*, 25(2), pp.333–358. doi:<https://doi.org/10.1016/j.jcps.2014.08.002>.

CMA (2024) Grocery Loyalty Pricing: Review of Supermarket Pricing Practices. London: Competition and Markets Authority.

Conly, S. (2012) *Against Autonomy: Justifying Coercive Paternalism*. Cambridge: Cambridge University Press.

Ebster, C. and Garaus, M. (2015) *Store Design and Visual Merchandising: Creating Store Space That Encourages Buying*. 2nd edn. New York: Business Expert Press.

Iyengar, S.S. and Lepper, M.R. (2000). When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, [online] 79(6), pp.995–1006. doi:<https://doi.org/10.1037//0022-3514.79.6.995>.

Kahneman, D. (2011) *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.

Kahneman, D. and Tversky, A. (2000). Prospect Theory: An Analysis of Decision under Risk. *Choices, Values, and Frames*, pp.17–43. doi:<https://doi.org/10.1017/cbo9780511803475.003>.

Andreu Mas-Colell, Green, J.R., Chiaki Hara, Ilya Segal, Tadelis, S. and Whinston, M.D. (1995). *Microeconomic theory*. Oxford: Oxford University Press.

Melé, D. and Cantón, C.G. (2014). The Homo Economicus Model. *Human Foundations of Management*, [online] pp.9–29. doi:https://doi.org/10.1057/9781137462619_2.

Pettinger, T (2019). Prospect Theory - Economics Help. [online] [Economicshelp.org](https://www.economicshelp.org). Available at: <https://www.economicshelp.org/blog/glossary/prospect-theory/>.

Popper, K. (1959) *The Logic of Scientific Discovery*. London: Hutchinson.

Scheibehenne, B., Greifeneder, R. and Todd, P.M. (2010). Can There Ever Be Too Many Options? a Meta-Analytic Review of Choice Overload. *Journal of Consumer Research*, 37(3), pp.409–425.

Simon, H.A. (1955). A Behavioral Model of Rational Choice. *The Quarterly Journal of Economics*, [online] 69(1), pp.99–118. doi:<https://doi.org/10.2307/1884852>.

Thaler, R.H. and Sunstein, C.R. (2008). *Nudge: Improving Decisions about Health, Wealth, and Happiness*. London: Penguin Books.

Tversky, A. and Kahneman, D. (1974). Judgment under uncertainty: Heuristics and Biases. *Science*, 185(4157), pp.1124–1131.